

4 Mechanisms Underlying Acute Changes in Range of Motion

Thixotropic Effects

In fact, it is not even necessary to stretch in order to temporarily increase range of motion (ROM). Warming the muscles and tendons will easily improve your elastic flexibility. Even more effective is to include an aerobic activity or warm-up muscle contractions. The traditional warm-up is typically initiated with a submaximal aerobic component (e.g., running, cycling) to raise the body temperature 1–2°C (121, 122). Any study that measures joint ROM before and after almost any activity that involves some persistent muscle contractions will detect an increase in ROM. S. Peter Magnusson: a well-respected stretch researcher from Denmark indicated that the acute effects of stretching in the holding phase of a stretch are due to changes in tissue visco-elasticity. The underlying mechanism for this visco-elastic effect is thixotropy. Thixotropy occurs when viscous (thicker) fluids become less viscous or more fluid-like when agitated, sheared, or stressed. When the stress is removed, or desists then the fluid takes a certain period to return to the original viscous state. Muscle contractions are not very efficient. Only 40–60 percent of the energy consumed during a contraction contributes to producing force (i.e., myofilament, Ca^{++} , and Na^+/K^+ pump kinetics), whereas 40–60 percent is released as heat (105). The muscle contraction-induced increase in temperature of the soft tissues can decrease the viscosity of intracellular and extracellular fluid providing less resistance to movement. Increases in muscle temperature can occur with higher environmental temperatures, and muscle contractions associated with dynamic stretching movements, isometric contractions during PNF stretching and to a lesser extent with the reflexive contractions of static stretching. Hamstrings passive static stretching elevated the local skin temperature by approximately 0.3°C after 180 seconds of stretching (32). Thixotropic effects on viscosity are not just muscle-related, but as tendons consist of 55–70 percent water (62), they will also be affected by viscosity changes.

A great analogy for those individuals who live in northern climates is the viscosity of the oil in the car or truck engine when it is extremely cold. If you imagine that the pistons are myosin molecules, both have to move in order to create movement. Oil in very cold temperatures would act more like viscous

molasses providing high resistance to the movement of the pistons. A cold muscle would have more viscous sarcoplasm providing higher resistance to the intramuscular proteins such as myosin, titin and others. The extracellular fluid would also be more viscous when cold and thus provide more resistance to the movement or sliding of muscle fibres, tendons and fascia. If a car is cold and will not start, you plug in the block heater which can warm up the oil (decrease viscosity) so the pistons can move. If your muscles are cold, muscle contractions can increase musculotendinous temperatures decreasing viscosity and resistance to movement. In fact, I have had lazy friends who rather than stretch and actively warm up before a squash game, would sit in the sauna to warm their bodies and thus “limber up” (i.e., decrease their musculotendinous and myofascial resistance to movement). Athletes playing in cold environments must keep this concept in mind. The colder they become, less pliability and less flexibility will result. Often, you may see or hear an athlete who will disdain from standing by a heater or using a blanket or jacket while on the sidelines in a cold or freezing environment. Their intended approach or message is that they are so psychologically “tough” they do not need to use the same devices as some “soft” or psychologically “weaker” opponents. As a scientist and a coach, I would rather that my athletes stay warm and be psychologically “smart” and physiologically “efficient”, as hypothermia (cold) not only affects resistance to movement (flexibility), but strength, power, rate of force development, endurance, metabolism, and other vital processes for success in the event or sport (36, 41).

There are a number of other explanations for the increased ROM immediately after stretching. Depending on whether the stretching is static, PNF, or dynamic, there may be various emphases on whether it is thixotropic, neural, mechanical, or psychological (stretch tolerance).

Neural Mechanisms of Acute Static Stretching

Although generally we try to relax when performing passive static stretches, there is still some muscle activation. Blazeovich et al. (19) recorded electromyography (EMG) activity (normalized to maximal voluntary isometric contraction – MVC) of 8.5 percent (soleus) and 5.5 percent (medial gastrocnemius) at the point of maximal stretch, thus suggesting that muscle reflexes would likely have contributed to the passive stretch resistance of the musculotendinous unit. They split their participants into flexible and inflexible individuals and found a moderate (0.60) correlation between the angle of EMG onset and maximum range of motion. Thus, more flexible people did not experience significant muscle activity (reflex-induced EMG) until they achieved a greater ROM. According to Nathalie Guissard and Jacques Duchateau (51), two internationally distinguished neuromuscular physiologists from Belgium, the amount of stretch or joint ROM that can be produced is highly attributable to the extent of muscle resistance caused by tonic reflexes. Whereas, dynamic movements (not through a full ROM) and

dynamic stretching (a full or nearly full ROM) tend to excite the neuromuscular system, static stretching is purported to decrease or disfacilitate this reflexive activity excitation of the motoneurons (11, 12, 13, 15). The origin of this reflex suppression can be tested by using the Hoffmann reflex (H-reflex) and the tendon reflex (T-reflex) (Figure 4.1). The H-reflex is evoked by stimulation of the Ia fibres to monitor the afferent excitability of the alpha motoneurons (38). Decreases in the H-reflex amplitude can signal decreases in motor neuron excitability or presynaptic inhibition (inhibition of interneuron (s) innervating Ia terminals) of the Ia afferents. What is the difference between inhibition and disfacilitation? Inhibition is a more active process caused by the activation of inhibitory interneurons leading to inhibitory postsynaptic potentials in the motoneurons (112). These interneurons would release inhibitory neurotransmitters such as γ -aminobutyric acid (GABA). However, with disfacilitation, motoneurons would be hyperpolarized due to the temporal absence of excitatory synaptic activity (112). In simpler terms, if you were riding a bicycle and you slowed your pedaling rate or stopped rotating the crank, you would decrease the activity and the speed of the bicycle would be disfacilitated. However, if you applied the brake pads to the wheels then you would actively attempt to slow the bike just like an interneuron and thus you would be directly inhibiting the bike's speed. The T-reflex could be a combination of inhibition or disfacilitation. T-reflex is elicited by tapping

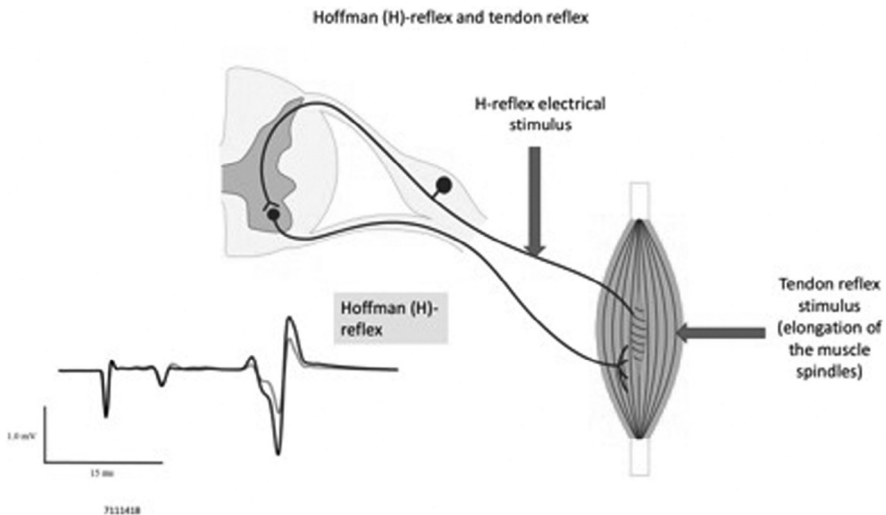


Figure 4.1 Hoffman (H-) reflex and tendon reflex. The stimulus for the H-reflex is a low-intensity electrical stimulation sufficient to activate the afferents to the spinal motoneurons but minimally activating the alpha motoneuron efferents so as not to elicit a strong muscle action potential (M-wave). The stimulus for tendon reflex is an elongation of the muscle to excite or activate the nuclear bag and chain fibres of the intrafusal muscle spindles to elicit a myotactic reflex

(percussion) of the tendon and could be affected by central changes (disfacilitation of motor neuron excitability and presynaptic inhibition) and by changes in the muscle spindle sensitivity/activity (disfacilitation). One hour of repeated passive stretching of the plantar flexors has been shown to decrease the stretch reflex by 85 percent while the H-reflex was reduced by 44 percent (7). However, you would have to ask, who would ever stretch their calf muscles for an hour. A good example how lab research which tries to understand mechanisms does not always parallel real life.

With passive static stretching, the muscle is typically extended at a slow to moderate rate into an elongated position by another person or a device (i.e., rubber band or machine) (see Figure 4.1). This extended or elongated position is then held for an extended period of time, typically from 15–60 seconds (3, 12). It is the responsibility of the muscle spindles to detect, continuously monitor and send signals to the central nervous system regarding the rate and extent of elongation of the muscle (80, 81). It can be regarded as both a protective system and a proprioceptive (position sense) system. As a protective system, if the muscle is elongated at a rapid rate to the end or near the end of the ROM then a reflexive signal will be sent to the spinal cord to contract the muscle that is being stretched (myotactic stretch reflex: see Figure 4.2) (82). This reflex is the action of the quadriceps muscle that you experience when a doctor taps your patellar tendon (front of the knee) and then your lower leg jerks forward (T-reflex). The doctor's hammer has elongated the patellar tendon which is attached to the quadriceps; stretching that muscle, which was detected by the muscle spindles leading to a reflex to the spinal cord, causes your

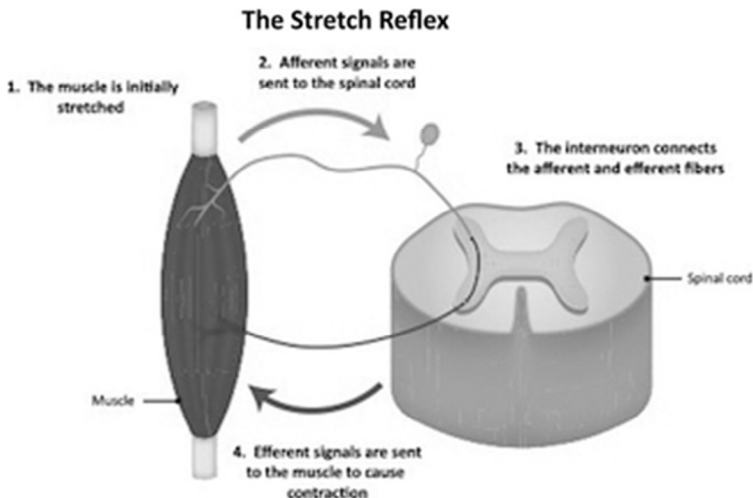


Figure 4.2 Tendon or myotactic reflex with monosynaptic excitation of the agonist muscle and di-synaptic reciprocal inhibition of the antagonist muscle (reciprocal inhibition not shown in this figure)

quadriceps to contract. This reflex contraction of the elongated muscle should therefore help prevent the joint skeletal structures (bones and cartilage) and ligaments from exceeding the joint and muscle's functional ROM and thus helping to prevent damage or injury. While the elongated muscle is being reflexively contracted the antagonist to that muscle is being inhibited. The myotactic reflex is monosynaptic (one synapse), whereas the antagonist inhibition is di-synaptic (two synapses). In terms of movement, this antagonist inhibition is efficient as it automatically reduces muscle contractions that work in opposition to the intended motion. Thus, if you are walking, the dynamic contraction and elongation of the quadriceps will lead to further excitation of the quadriceps, which are propelling you along while the hamstrings are being inhibited to reduce the resistance to the walking movement. However, if you are sprinting and the hamstrings get stretched or elongated to a great extent, they also will be subjected to myotactic reflexes to protect them from being extended too far.

Within the muscle spindle, there are nuclear chain fibres that preferentially respond to changes in the amount of stretch or elongation of the muscle whereas the nuclear bag fibres respond to both the extent and rate of elongation (see Figure 4.3). When activated, nuclear chain and bag fibre impulses are sent to the spinal motoneurons via annulospiral (high conduction velocity) and flower spray endings (low conduction velocity) through Ia (nuclear bag and chains) and II (nuclear chains) afferents respectively. As previously mentioned, a monosynaptic (just one synapse) myotactic reflex is initiated from the rapid stretch action resulting in the depolarization/activation (contraction) of the alpha motoneuron of the stretched muscle and inhibition (di-synaptic) of the antagonist muscle (82).

The spindles also work to inform the system about a body segment's position in space or proprioception. The intensity or discharge rate of the impulses from the muscle spindles will inform the central nervous system about the

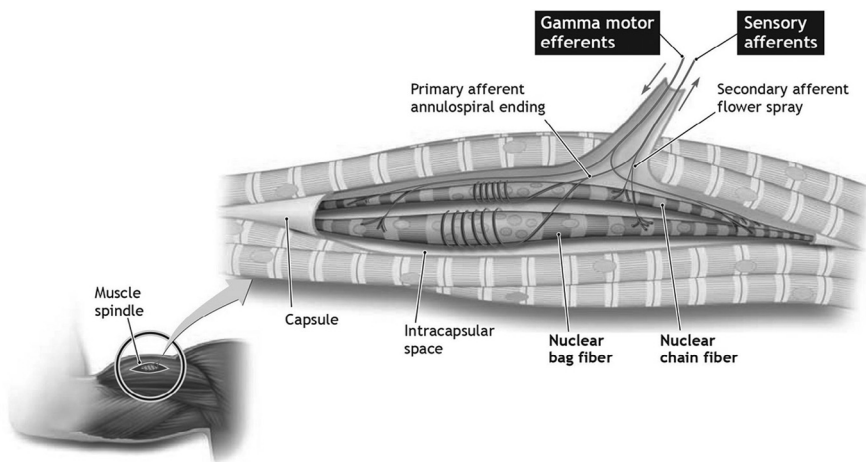


Figure 4.3 Nuclear bag and chain fibres are located within an intrafusal muscle spindle

rate and extent of movement (97). That is why you can close your eyes, abduct and extend your arms and then still touch your nose. Your muscle spindles are informing the system about how fast and far you have moved in order to reach your nose and based on prior experience, when to slow down and stop so you do not break your nose.

Another receptor that is activated by tension exerted on the muscle tendon and muscle is the Golgi tendon organ (GTO) (26, 57). The GTO receptors within the tendon are shaped like coiled elastics and when they are uncoiled due to tension, a piezo-electric effect occurs (see Figure 4.4). Certain biological materials such as crystals (bone), proteins, DNA and others can discharge an electric charge in response to mechanical stress (102). Piezoelectricity means electricity resulting from pressure, derived from the Greek *piezō* or *piezein*, which means to squeeze or press. So, when a contraction occurs or

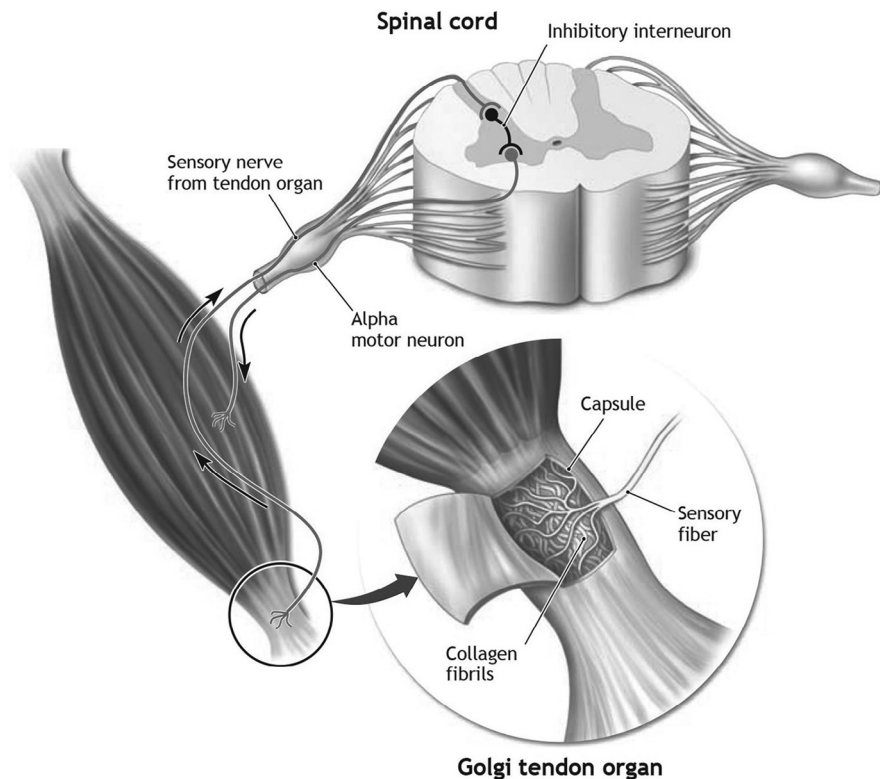


Figure 4.4 Golgi tendon organs (GTO) are located in the muscle-tendon or aponeurosis-tendon junction and they can lie in series or in parallel with the extrafusal muscle fibres, so they can detect force increases not only from fibres in series with the receptor but distant to the GTO through the parallel connections

tension is exerted on the musculotendinous unit, the GTO discharges signals to the central nervous system. While the GTO can reflexively excite the system, under these circumstances, it typically results in an inhibitory signal that is labelled as an autogenic inhibition. This reflex is di-synaptic in that the GTO synapses with an inhibitory interneuron in the spinal cord, which then inhibits the same muscle group that experienced the tension (61). Although theoretically, the GTO should contribute to stretch-induced reflex inhibition, Edin and Vallbo (37) found that while most spindle afferents respond rapidly to stretch, GTO were insensitive to the tension produced on the tendon with stretch. If there is a stretch-induced GTO inhibition, it is more likely to occur with large amplitude stretches (51). Furthermore, any possible GTO inhibition subsides almost immediately (60–100 milliseconds post-stretching) after the stretching discontinues (57).

In conjunction with the GTO inhibition to large amplitude stretches, the Renshaw cells can also play a minor role with large amplitude stretches. Renshaw cell inhibition is also known as recurrent inhibition (see Figure 4.5). Recurrent or Renshaw cell inhibition can exert stabilizing effects on motoneurone discharge variability, and motor unit synchronization during voluntary muscle contractions (79). Recurrent inhibition is more prevalent with weak rather than strong contractions, and phasic rather than tonic contractions (59). As the stress or tension

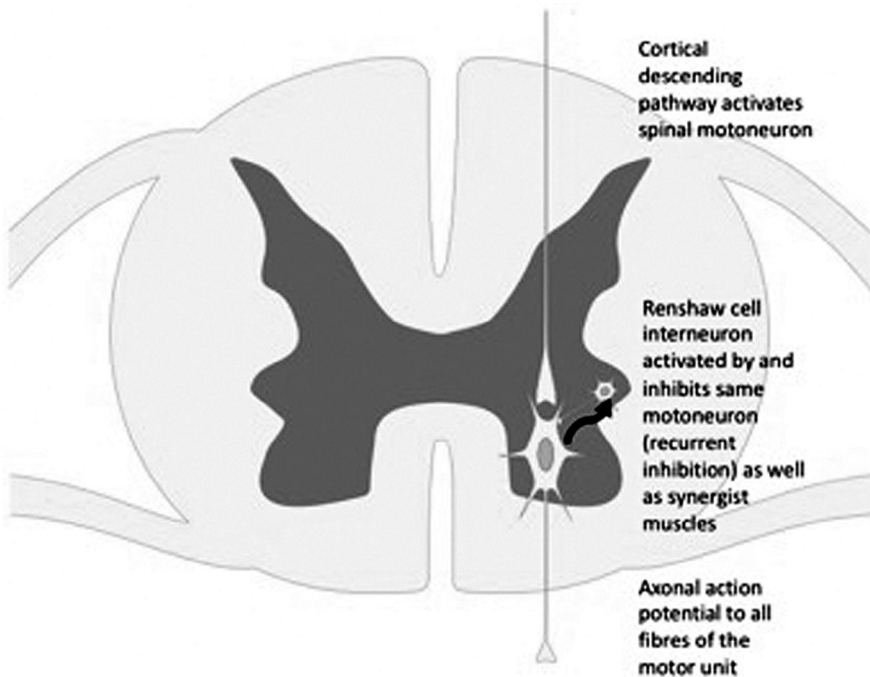


Figure 4.5 Renshaw cells and Recurrent Inhibition

on the muscle during a large-amplitude stretch is still much less than with a maximal contraction, it should contribute to motoneuron inhibition and muscle relaxation. However, as it is more prevalent with phasic contractions, it may play a more predominant role with full ROM dynamic stretching compared with lesser effects with static stretching.

We have discussed the possible roles of nuclear chain and bag fibres of the muscle spindles, Renshaw cells, and GTOs. Cutaneous nerve fibres can also contribute to the ROM capabilities (see Figure 4.6). Cutaneous receptors have polysynaptic innervations to motoneurons and are monitored with the exteroceptive (E-) reflex. Research has demonstrated that small amplitude passive stretches induce pre-synaptic inhibition (e.g., decreased H-reflex) but no change in the E-reflex (cutaneous) (34) or corticospinal excitability as measured with motor evoked potentials induced by transcranial magnetic stimulation. However, with a greater ROM, both H- and E-reflexes decreased similarly, suggesting that post-synaptic inhibitory mechanisms contribute to the observed changes, but they only persist for a few seconds following the completion of the stretching. According to Guissard and Duchateau, (51), joint and cutaneous receptors are not significant inhibitors with small amplitude stretches with only a small contribution during large amplitude stretches. Furthermore, increases in the afferent excitability of the motoneurons (H-reflex) are associated with increases in the musculotendinous passive torque (31).

With an active static stretch, the individual will use another muscle group – typically, the antagonist muscles to help move the target muscle and joint through an extended ROM. For example, to stretch the hamstrings, you may

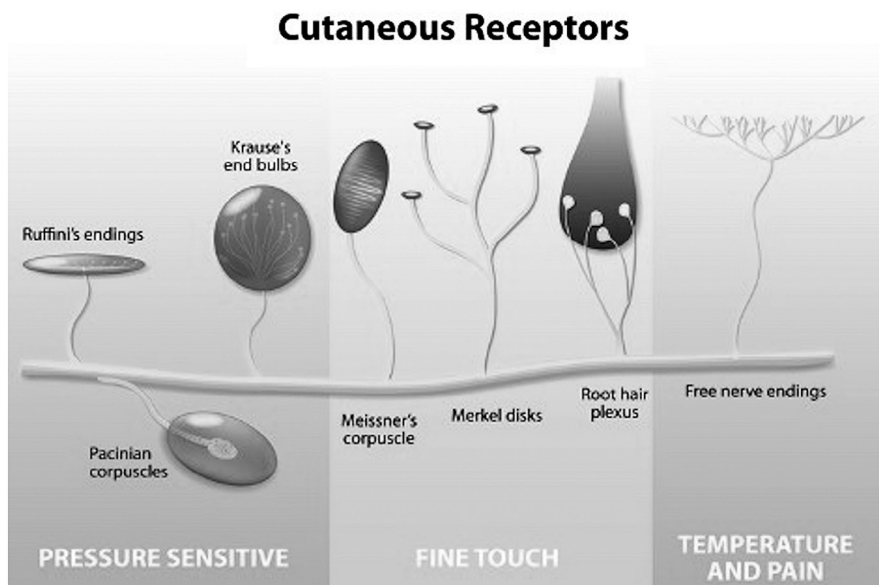


Figure 4.6 Cutaneous nerve receptors

actively contract the quadriceps to flex the hip which will naturally extend the hamstring muscles which work as hip extensors. Reciprocal inhibition (29, 95) may play a role in this situation, as the quadriceps contraction changes the length of the quadriceps muscles spindles at a particular rate and can lead to an inhibitory di-synaptic reflex synapse with its antagonist, leading to more relaxation in the hamstrings.

So, let's go through a static stretching scenario of a person who is lying on their back (supine position) and in the first instance, a partner flexes the hip of their extended leg towards their chest to stretch the hamstrings and in the second situation the person contracts their quadriceps and uses their arms to pull their extended leg toward their chest. In the passive static stretching situation of the first scenario, the muscles spindles, specifically, the nuclear bag fibres would detect a slow to moderate rate of muscle length change, while the nuclear chain fibres would predominately monitor the extent of change in muscle length. Both nuclear bag and chain fibres would discharge at a relatively high frequency with annulospiral and flower spray ending through the Ia and II afferent nerve trunks to initiate the myotactic reflexes and cause contractions of the hamstrings, thus resisting the stretching of the muscle. In one study, although subjects were instructed to relax during a dorsiflexors stretch, EMG activity reached 5.5 percent and 8.5 percent of MVC in the gastrocnemius and soleus muscles respectively (19). Therefore, muscle spindle reflexes would have contributed to the "passive" resistance of the musculotendinous unit during the stretch.

Naturally, once the stretch had reached the maximum ROM, there would no longer be any change in the rate of stretch. Within seconds, the nuclear bag fibres would begin to decrease their discharge rate, hence decreasing the intensity of the myotactic reflex contractions (disfacilitation). As the stretch was held in this position for a prolonged period (typically 15–60 seconds), the nuclear chain fibres would accommodate this new position and also decrease their discharge frequency (disfacilitation). This decreased discharge frequency can partially be attributed to the actions of the gamma efferent system (see Figure 4.3). The gamma efferent system attempts to return the muscle spindles to their reference length after or during movement (81). Just like the extrafusal muscle (skeletal muscle innervated by the alpha motoneuron), the intrafusal muscles (location of sensory muscle spindles) also have contractile myofilament proteins. Thus, the gamma efferent system can activate these myofilaments leading to a spindle contraction that would shorten the length of the spindle. By decreasing the length of the spindle, the nuclear chain fibres decrease their discharge rate and the myotactic contractions decrease further, leading to more relaxation of the muscle. Although elongating the muscle to the end of ROM would have placed tension on the GTO that should activate greater type Ib afferent inhibitory stimulation there is little evidence that it contributes to substantial muscle relaxation (perhaps a small contribution with an extensive ROM to maximal point of discomfort). Try this yourself. Stretch any muscle passively and hold it for 30 seconds or more. Then see if

you can stretch the muscle or increase the ROM further. It is guaranteed that you will have a greater ROM. If you used an active static stretch then the contraction of the quadriceps may have also contributed to further relaxation with the reciprocal inhibition effect.

So, which process is more predominant with static stretching: inhibition or disfacilitation? In one study H- and T-reflexes were compared in the soleus with increases in ankle ROM (52). The H-reflex amplitude decreased more than the T-reflex (31 percent vs. 8 percent of control). When the stretching was completed the H-reflex returned immediately to its control value, while the T reflex remained depressed. Therefore, the T-reflex reduction was more likely derived from either a decrease in muscle spindle sensitivity (disfacilitation) or increased musculotendinous unit compliance (mechanical change). According to this study, during the stretch there could be pre-synaptic inhibition and muscle spindle-induced disfacilitation of the motoneuron, whereas the persistent increases in ROM would be attributed more to spindle disfacilitation and perhaps musculotendinous compliance. H-reflex inhibition has been reported to have two recovery stages during 30 seconds of stretching. It was hypothesized that H-reflex inhibition present until 18 seconds was related to post-activation depression of Ia afferents caused by the passive dorsiflexion movement. From 21 seconds, the lower inhibition may have been caused by a weaker post-activation depression, and inhibition from secondary afferents or post-synaptic inhibition (23).

Whilst the H-reflex can be inhibited during the stretch, this H-reflex inhibition is short lived following the stretch (22). This rapid recovery has been suggested to be influenced by a transient increase in corticospinal excitability (22). Further evidence for increased cortical excitability was provided by Opplert et al. (94) who found increased corticospinal excitability but a lack of change in spinal excitability after both static and dynamic muscle activity.

Budini et al. (24) reported that tendon tap reflexes could be inhibited for ten minutes after 30 seconds of static stretching. They demonstrated that it was possible to promote an immediate recovery from this spindle inhibition or disfacilitation and restore reflex sensitivity by performing a maximal voluntary isometric contraction. However, three dynamic contractions delayed the spindle reflex recovery by 90 seconds.

In fact, we can go even deeper than that. Are there differences in how the muscle reflexes are inhibited if you have a small versus large amplitude stretch? It has been shown that passive small amplitude stretches decrease the H-reflex through pre-synaptic inhibition. Pre-synaptic Ia inhibition pathway originates from the intrafusal muscle spindle fibres and projects pre-synaptically (before the junction of the dendrite and the motoneuron) by interneurons onto the Ia terminals (53). Thus, there is no intrinsic change in motoneuron excitability. With larger amplitude passive stretches, there may be a suppression of the cortical neurones and $\downarrow$ motoneuron excitability (53). In this case with large amplitude stretches, GTO inhibitory afferents could contribute to decreased motoneuron excitability from Ib fibres (53). GTO respond primarily to muscle contraction forces, but are not very sensitive to the degree of

force or stress associated with smaller amplitude passive stretching (53). Remember though, that GTO effects would only continue during the period of stretch and tend to diminish rapidly thereafter (57). There is also the possibility of post-synaptic inhibition with large amplitude stretches from the inhibitory effect of the Renshaw cell recurrent inhibition loop (25).

In fact, it is possible, that stretch-induced reflex interneuron inhibition might affect not only the muscles being stretched but distant or non-local muscles as well. There are a few studies that have demonstrated that stretching the hamstrings unilaterally (only one side of the body) will improve ROM of the contralateral (opposite side) hamstrings (27). Our lab has also shown that static or dynamic stretching of the shoulders will increase ROM of the hamstrings, while stretching the hip adductor (groin) muscles increased the flexibility of the shoulders (14). This research could be crucial evidence for the pervasive effects of stretch-induced neural inhibition that acts globally on the body.

Neural Mechanisms of Acute Proprioceptive Neuromuscular Facilitation Stretching

Traditionally, Proprioceptive Neuromuscular Facilitation (PNF) was considered more effective than static stretching for increasing ROM due to the variety of reflex inhibition techniques that were proposed to be involved. For example, with the CRAC (contract-relax-agonist-contract) method, the muscle to be stretched (e.g., hamstrings) would be actively placed in an elongated position by contracting the antagonist muscle (e.g., hip flexion by the quadriceps). The contraction of the antagonist would activate reciprocal inhibition. Reciprocal inhibition is a reflex branch of the myotactic reflex. As a reminder, the myotactic reflex is a monosynaptic reflex initiated from the stretching of the muscle spindles resulting in an excitation of the stretched muscle. This muscle spindle-induced excitation also has an afferent nerve branch that excites an inhibitory interneuron that suppresses the activity of an antagonist motoneuron (di-synaptic). In this example, contracting the quadriceps to lengthen the hamstrings would change the length of many quadriceps muscle spindles (especially in the distal regions), causing further excitation of the quadriceps with inhibition of the hamstrings permitting greater muscle excursion (elongation). The next step with CRAC PNF stretching would be to contract the stretched muscle (e.g., hamstrings) against a partner's resistance or it can also be against an immovable object like a wall. This contraction, which does not need to be a maximal contraction, as Roger Enoka (a very prominent and internationally renowned neuromuscular scientist) and colleagues (38) found that whether the contraction was performed at 50 or 100 percent of the subjects' maximal voluntary force, there was a similar reduction in the H-reflex amplitude (afferent excitability of the alpha motoneuron). Furthermore, this contraction of an elongated muscle would place stress on the GTO, possibly leading to its activation inducing autogenic inhibition (56). Autogenic inhibition from GTO activation increases Ib muscle afferent activity (57). Ib afferent activity hyperpolarize the

dendrites synapsing with spinal alpha motoneurons of the stretched muscle, decreasing or blocking the Ia afferent reflex activity enabling further increases in ROM (26). However, there is no direct evidence for a positive relationship between the aforementioned reflex activity and PNF ROM (26). One might expect that this reciprocal and autogenic reflex inhibition would decrease muscle activity as evidenced by attenuated EMG activity, but a number of studies have illustrated increased resting EMG activity immediately after the contraction phase of a PNF stretch (71, 87). As GTO effects persist briefly after the tension to the receptors is removed (113), the autogenic inhibition effects should not persist and its contribution to PNF flexibility is highly debatable (56, 103). Although, a depression of the spinal reflex excitability after the isometric contraction is brief (<5 seconds), it could still persist long enough to provide an advantage for the subsequent stretch. Thus, similar to static stretching, there could be some reflex inhibition during the stretching procedure. An increased stretch tolerance (86), decreased visco-elasticity and a degree of reduced musculotendinous stiffness (60, 71) could all contribute to the sustained increase in elastic ROM.

Neural Mechanisms of Acute Dynamic Stretching

Whereas static and PNF stretching should reduce muscle activation through some degree of reflexive disfacilitation with reduced nuclear bag and nuclear chain muscle spindle receptor activity (Ia afferents) with further possible inhibition from autogenic (Ib afferents) and reciprocal inhibition (Ia afferents), dynamic stretching should excite or increase activation of the system. The previously discussed myotactic reflex activity would be increased with dynamic stretching due to nuclear bag and chain excitation from the higher rate and extent of muscle elongation respectively. As dynamic muscle stretching is usually performed as a relaxed action with submaximal muscle contractions, the activation of GTOs initiating autogenic inhibition would not be expected to play a major role in increasing acute ROM. Reciprocal inhibition would be activated with the sequential movement of the limbs similar to the well documented reciprocal inhibition sequences found with locomotor activities such as walking and running (96, 98). This reciprocal inhibition could contribute to greater dynamic movement excursions during the stretching activity but would not persist after the activity. Hence, for the augmentation of ROM to persist after the dynamic stretching, the reciprocal inhibition would need to contribute to viscous and morphological changes, that would continue for a prolonged period after the stretching. Typically, dynamic stretching is not as uncomfortable or painful as static or PNF stretching, hence the role of stretch tolerance may not be as predominant.

Stretch Tolerance Sensory Theory

Increases in ROM could also be attributed to a psycho-physiological effect (sensory theory). Magnusson and colleagues suggest that increased flexibility can be largely attributed to an increased stretch tolerance (72, 73). They showed that after three weeks of stretch training, the increased ROM was not attributed to differences in muscle stiffness or reflex EMG activity (72). Increased stretch tolerance could be related to changes in the sensitivity of nociceptive (pain) nerve endings (77) that allow the individual to accommodate greater discomfort or pain and thus push themselves through a greater ROM. Stove and colleagues (106) induced remote pain (cold pressor test of non-dominant hand) intended to elicit endogenous pain inhibitory systems and found an increase in knee extension ROM. They suggested there was a link between increased stretch tolerance and endogenous pain inhibition. Diffuse noxious inhibitory control theory indicates that activity of A delta- or A delta- and C-peripheral fibres influence convergent neurons in the spinal dorsal horn resulting in a global release of endorphins and enkephalins (morphine-like compounds) that promote analgesic effects allowing the individual to sustain greater discomfort or pain and stretch further (65). Enhanced stretch tolerance rather than changes in passive muscle stiffness has been demonstrated with seniors regardless of sex (91). Just 20 seconds of static stretching increased ROM but did not affect the shear elastic modulus (muscle elasticity) again pointing to an increase in stretch tolerance as the mechanism (100). Their ROM improvement persisted for 10 minutes. Passive stretching can produce acute ischemic compression, which has been shown to result in reduced perceived pain in the neck and shoulder muscles (54). Freitas et al. (45) in a meta-analysis examined 26 stretch training studies that ranged from 3–8 weeks in duration with a weekly stretching duration of 1,165 seconds per week (about 20 minutes). Stretching for 3–8 weeks did not on average alter muscle and tendon properties and thus the increased extensibility must have been related to a greater tolerance to tension. However, if stretch tolerance was the only mechanism at play with no substantial reflex inhibition occurring, then the commonly found static stretch-induced performance impairments would not be expected from non-stretched muscles.

But we found that not only can stretching one muscle increase the ROM of another muscle but we have also reported that unilateral static stretching of the plantar flexors (calf muscles) led to impairments of jump height in the contralateral lower limb (30) and stretching the shoulders also impaired lower limb jump performance (75). Static stretching of the pectoralis major muscle decreased activation of the triceps brachii during a bench press action (76). There are not many questions in the world that are simply black and white. So, while an increased stretch tolerance (totally psychological or psycho-physiological influenced by type III and IV nociceptor afferents) can certainly help explain ROM improvements, it seems likely that some degree of afferent stretch-induced reflex inhibition is also acting on the muscle under stretch as

well as other non-affected muscles. While this reflex inhibition would be most predominant during the holding of the stretch, the reported subsequent impairments in non-local jump performance and muscle activation suggests that it can persist for a few minutes following stretching. However, even, if neural reflex inhibition works primarily during the actual stretching exercise permitting a greater elongation of the muscle, then it might impact morphological structures such as muscles, tendons and in some cases ligaments.

Acute Morphological Static Stretching Mechanisms

Skeletal Structures

Greater stretch tolerance or neural inhibition should allow the muscle to be elongated to a greater degree. Maintaining this greater elongation over an extended period (i.e., 20–60 seconds) might be expected to affect the properties of the musculotendinous tissues. What musculoskeletal components restrict our ROM. ROM is affected by skeletal structures, joint capsules, fascia, ligaments, muscles, tendons, aponeuroses, and fat. What factors can we modify with stretching? If we stretch till a bone is fractured, we can get an acute increase in ROM but at what cost: excessive pain, inflammation, and loss of function. Within moments, the pain and inflammation will then decrease the ROM! The glenoid fossa or cavity of the shoulder (glenohumeral) joint is a relatively flat surface allowing the shoulder a great deal of unrestricted movement for flexion/extension, abduction/adduction, horizontal abduction/adduction, medial and lateral rotation, and circumduction (see Figure 4.7). In contrast, the acetabulum of the hip joint is deeper and more cup-like restricting ROM compared

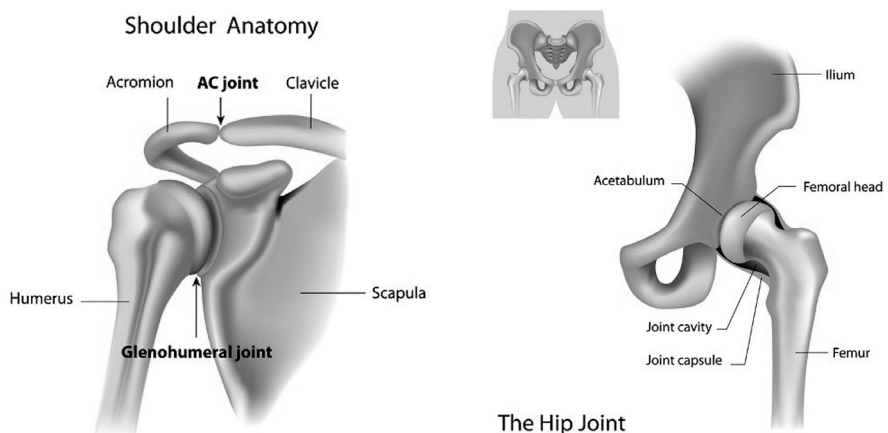


Figure 4.7 Skeletal restrictions on ROM. Hip acetabulum has a much deeper cup-like structure versus the flatter more mobile shoulder glenoid fossa

with the glenoid fossa (see Figure 4.7). It can also perform flexion/extension, abduction/adduction, medial and lateral rotation and circumduction. Whereas, shoulder motion is expansive, hip ROM is quite limited compared with the shoulder.

<i>Normal ROM (degrees)</i>	<i>Hip</i>	<i>Shoulder</i>
Flexion	110–120	180
Extension	10–15	45–60
Abduction	30–50	150
External rotation (lateral)	40–60	90
Internal rotation (medial)	30–40	70–90

Thus, skeletal structures are typically not modifiable with any amount or type of warm-up and stretching will typically not affect the skeletal structures.

Ligaments

Ligaments are tough, fibrous tissue that connect one bone to another bone providing joint stability. An important function is to prevent movement that might damage a joint. In most circumstances, the objective of stretching is NOT to elongate the ligaments. Lengthening ligaments would decrease joint stability and often lead to injury. There are some cases however, when the goal of stretching would be to elongate ligaments. Athletes who need to attain extreme levels of flexibility such as gymnasts, figure skaters, dancers, certain circus performers and others may target their ligaments.

Some individuals have a predisposition to hyper-laxity of their connective tissue or hypermobility of their joints. Women tend to have greater joint laxity or hypermobility compared with men (16, 18, 33, 63, 120). Harry Houdini (1874–1926) was famous for his escape abilities (see Figure 4.8). He could be tied up in ropes or chains and submerged in water or other hazards (i.e., suspended from buildings) and would make miraculous escapes. Houdini had lax ligaments and could subluxate (partially dislocate) or dislocate many of his joints. Hence, movements that were impossible for the average individual, were possible for Houdini by voluntarily subluxating a joint, sliding or moving out of a chain or rope and then popping the joint back into place before reappearing before the audience. Naturally, you should not need to be warned that you should not try this at home. Average individuals who experience an injury with a joint dislocation or subluxation often have difficulties throughout their life with subsequent dislocations. People with hypermobile joints (see Figure 4.9) can experience increased chances for nerve compression disorders (99), impaired proprioception (10, 74) increased risk of joint trauma and osteoarthritis (17, 43, 49). Ligaments lack the extensive vascularity of muscles and tendons and tend not to heal or recover completely back to their original length and tension. So, unless there is a need for



Figure 4.8 Harry Houdini



Figure 4.9 Contortionist with hypermobile joints

extreme ROM capabilities by an athlete, stretching for health and normal function should not be so severe as to lengthen the ligaments (includes joint capsule). However, placing mechanical stress on ligaments as with stretching can play an important role in the processes of cellular differentiation up-regulating (increasing the activity of) ligament fibroblast markers such as collagen types I and III and tenascin-C (4). Thus stretch-induced mechanical stress would help build the ligament matrix making it stronger and more resistant to damage or injury. Similar effects are seen with tendon tissue, where a lack of mechanical stress can cause a progressive loss of cell matrix and a prolonged period of stress deprivation results in even higher mechanical stresses needed to promote tendon tissue growth than before the removal of stress strain (6).

Adiposity

Excessive fat can affect flexibility tests. An abundance of adiposity around the waist and trunk will impede hip flexion for example. But there is no concrete evidence that fat will impede the extensibility of tissues. An early study by (114), did not find any significant correlation between low fat, high fat and muscular individuals and their flexibility. Alter (3) points out that very large sumo wrestlers with an overabundance of fat can still exhibit extraordinary levels of flexibility. Of course, fat is not a tissue targeted during stretching during a warm-up. Proper diet and exercise are the key to improving fat-induced restrictions on some ROM tests (i.e., sit and reach or toe touch).

Nerves

When stretching, can we elongate the nerves? Nerves do have the capacity to elongate to a certain point. The extensibility of the nerve resides mostly with the perineurium (connective tissue sheath surrounding a bundle (fascicle) of nerve fibres within a nerve) with an elongation range between 6–20 percent of its resting length (3, 107, 108). Once the maximum length is reached, the nerve is susceptible to tearing or injury. If the perineurium or nerve is ruptured, there can be leakage of proteins into the fascicles and edema and reduced possibility for regeneration (50). Furthermore, stretching the nerve as much as 8 percent can reduce blood flow with complete occlusion at 15 percent elongation (67, 68, 93). Nerve conduction can also be inhibited with only 6 percent elongation (50, 115). Stretching to induce neural tension (four x 1-minute stretches of the hamstrings with head, neck and trunk slumped down (partially flexed)) contributed to knee flexion strength loss (84). As long as the maximum elongation limit is not exceeded then the nerve should be able to return to its original length (109). With muscle fascicles able to elongate approximately 50 percent, how is it possible that we can stretch without always damaging our nerves? Although the nerve itself can only elongate 6–20 percent, the nerves are not typically situated in a straight line but rather are in a slacker position due to an undulating path through the tissues (fasciculi) (104). So, when stretching, the nerve does not initially elongate,

but it actually straightens out till there is no longer an undulating course. Thus, the we can stretch far greater lengths than just 20 percent of the resting length of the musculotendinous tissues without damaging the nerves. Another evolutionary safety aspect is that most nerves traverse the flexor side of the joint. When a joint is flexed, the nerves would actually be placed in a more relaxed position rather than under the stress of an elongated position. Exceptions to this general rule are the ulnar nerve which traverse the extensor aspect of the elbow. You will experience the anatomical position of your ulnar nerve when you hit your “funny bone” and feel the “humour” of the pain running up your arm, as you hit the ulnar nerve near the humerus.

The sciatic nerve is another exception to the flexor positioning of nerves, as it runs across the extensor aspect of the hip. The sciatic nerve is protected by an especially thick epineurium, which constitutes about 88 percent of its cross-sectional area (108), as we spend an inordinate amount of time squatting (elongating the nerve tract) and sitting on the sciatic nerve (3). Although our primary focus is not stretching of the nerves, it has been shown that acute maximal dorsiflexion ROM can be increased by stretching the sciatic nerve without altering the muscle stiffness (5). The decrease in sciatic nerve stiffness was significantly correlated with the increase in maximal dorsiflexion ROM ($r = 0.571$). Therefore, while it is not a primary aim of stretching to elongate our nerves and associated connective sheaths, there can be a minor contribution to increasing ROM.

Muscle Hypertrophy

In the past, strength training resulting in muscle hypertrophy was feared to result in “muscle boundness”. Being “muscle bound” inferred having large muscles that would inhibit flexibility. Contrary to this myth, there are examples of hypertrophied bodybuilders such as “Flex Wheeler” who could perform the Russian splits (legs spread laterally or fully abducted) during competitions. There are many other hypertrophied athletes such as American football players, rugby, ice hockey players, sumo wrestlers, and others who exhibit superb levels of flexibility. Although performing partial ROM (also known as cheat “reps”) resistance training can shorten a muscle inhibiting ROM, there are a number of studies (66, 78, 101, 118, 119) and a recent meta-analysis (2) demonstrating increased ROM when performing full ROM resistance training. These resistance training ROM improvements are also seen with senior adults (9, 40). Although not every study finds an increase in flexibility with resistance training, there is typically no loss of ROM (48). Thus, large muscles have a great propensity for extensibility.

Intracellular and Extracellular Connective Tissue

Muscles have both intracellular and extracellular components. When stretching a muscle, what parts are actually being stretched? Myofibrils can be stretched to twice their resting length without damage (47). The elasticity of the myofibrils is largely due to an intracellular protein called titin, with a length of

approximately half the sarcomere being anchored at the Z disc (through a telethonin protein) and the M line (see Figure 4.10) (47). Titin can act as a spring that unfolds in response to high tension or stress. Myofibrils are connected together by the extracellular matrix. An essential protein in the extracellular matrix is fibronectin, which has elastic properties. Fibronectin can stretch up to four times its resting relaxed length. Fibronectin is composed of three subunits (FN-I, FN-II, and FN-III) with FN-III as the subunit that unfolds to contribute to the elasticity (see Figure 4.11). Stressing these units either mechanically, thermally or chemically leads to further binding with other fragments (e.g., anastellin) to form superfibronectins with increased adhesion capabilities. Fibronectins are also in contact with integrin molecules on the muscle membrane (47). Integrin molecules also sense membrane tension and transfer this information to the Raptor-mTOR complex in the nucleus helping to promote greater protein synthesis (58, 123). Specifically, integrin links laminin in the extracellular matrix with the cell cytoskeleton and translates mechanical forces into chemical signals. Stretching and muscle contractions activate intracellular signaling molecules that respond to injury-induced damage. Integrin stabilizes the muscle and provides communication between the matrix and cytoskeleton. Thus, elevated tension from stretching detected by the integrin molecule transduced to the nucleus leads to increased protein synthesis and helps with injury protection and prevention (21).

This enhanced muscle protein synthesis would increase muscle volume and strength, protecting it from other similar force stressors. An animal experiment

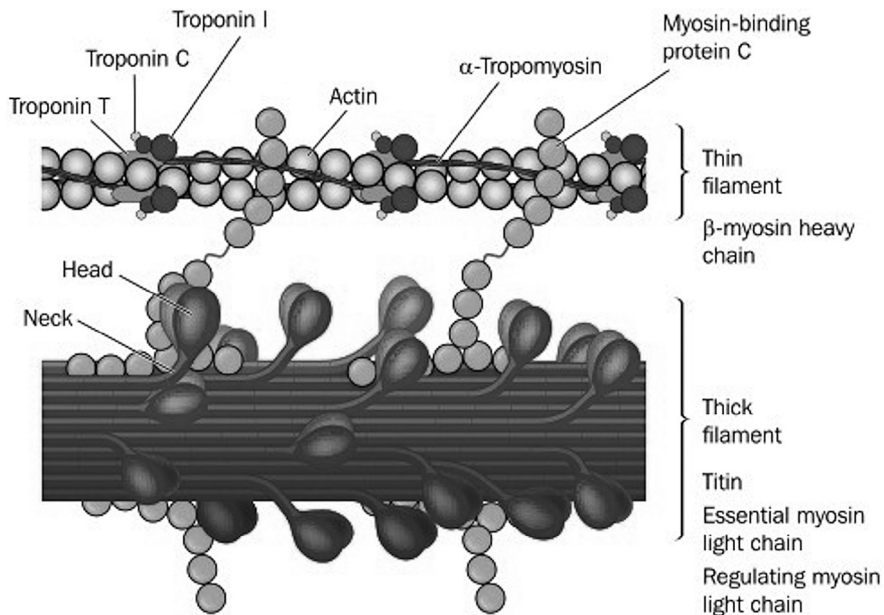


Figure 4.10 Titin and myofilament structure

Cell Adhesion Molecules (CAMs)

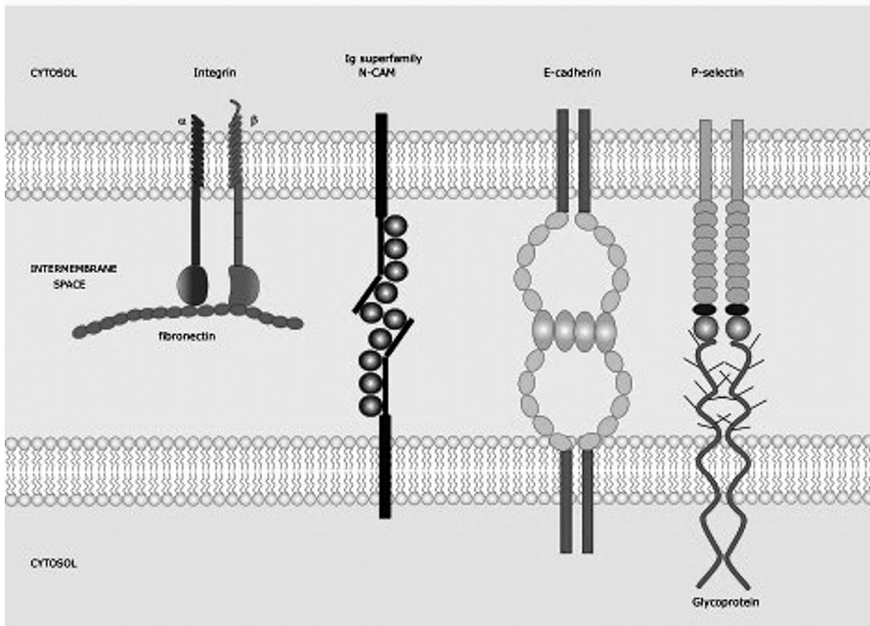


Figure 4.11 Fibronectin and integrin

had chicken latissimus dorsi (lats) muscles passively stretched over an extended duration (days). Fortunately, or unfortunately for the chickens, they were sacrificed and the muscles were used in isolation in a physiological saline bath solution. Researchers found an increase in muscle protein synthesis activity just by passively stretching the isolated muscle (64). Not all animal model experiments transfer easily to humans, and I would not suggest hanging from a bar or being placed on a rack for days or weeks to increase the hypertrophy of your “lats”. However, it does demonstrate how the tension from a passive stretch is transduced (signal is sent) by the membrane integrin molecule to the Raptor-mTOR complex in the nucleus to activate the transcription and translation activities of the DNA and DNA respectively to promote muscle growth (increased protein synthesis). This stretch-induced adaptation may contribute to the increased tolerance of the muscle matrix to external forces and torques decreasing injury incidence. We will discuss in-depth in another chapter, the evidence for stretch-induced increases in muscle strength and hypertrophy.

Tendons

Static stretching has often been prescribed as part of a warm-up or a training routine, as it is purported to decrease the incidence of musculotendinous

injuries (12, 83). The rationale for decreased musculotendinous injuries would be that the muscles and tendons would be better able to withstand or absorb the forces placed on the tissues and not tear (muscle strain or tendon sprain). Logically, then, increased ROM should not only be ascribed to decreased viscosity, neural inhibition, and stretch tolerance but also decreased stiffness or increased compliance of the muscles and tendons. There is much conflict in the literature as to the extent of tissue compliance/stiffness changes. Some researchers claim an association between reductions in myotendinous stiffness and ROM following an acute bout of static stretching (89), suggesting altered muscle mechanical properties are important contributors (89). Two very important stretch researchers, Anthony Blazeovich (Australia) and Anthony Kay (UK) investigated the contribution of muscle and tendon elongation to maximum dorsiflexor's ROM. They found that muscular tissue is more compliant than the tendon during passive stretch. In their study, muscle lengthened 14.9 percent compared with 8.4 percent lengthening of the tendon under maximal stretch conditions (19). Thus, the maximum dorsiflexion ankle ROM was not constrained by tendon elongation because further tendon strain should have been possible. Similarly, Abellaneda, (1) from the Guissard and Duchateau lab, indicated that the relative contribution of muscle fascicles to MTU elongation was 71.8 percent, whereas the tendon contributed 28.2 percent. Tendons are composed of 60–85 percent collagen protein (62), whereas connective tissue in muscle constitutes between 1 and 10 percent. Collagen is a very strong fibrous tissue explaining the decreased tendon compliance. It is important to note that these studies show the response of muscle and tendon to stretching. However, prior to stretching, the resting length of the tendon far exceeds the muscle fascicles and thus the tendon can account for about half of the change in musculotendinous unit length. Because tendon elongation exceeds muscle, then stress-induced muscle strain surpasses tendon strain by about four fold (70). Freitas et al. (45) in their meta-analysis of 26 stretch training studies found that there were trivial effects of stretch training on tendon stiffness whereas one study actually demonstrated lower tendon extensibility after three weeks of stretch training (20). Hence, while tendons contribute greatly to ROM, the lesser elongation of the muscle fascicles causes them to receive more of the strain caused by stretching, thus leading to a greater chance for muscle strains versus tendon sprains.

Fascicles

Another means to increase muscle length might be to change both the length and the angle of the muscle fascicles (fibres). In that same study by Blazeovich et al. (19), they reported that the contribution of changes in fascicle angle to the maximum stretch capabilities was negligible compared with the contributions of fascicle elongation. When comparing highly flexible gymnasts to volleyball players, rhythmic gymnasts displayed longer muscle fascicles at rest and greater fascicle elongation (distal part of medial gastrocnemius)

(35). There is also a minor contribution to increased muscle elongation from fascicle rotation. A significant difference between flexible and inflexible individuals is that fascicle rotation during stretch was greater in flexible subjects (~40 percent at 30° dorsiflexion) than inflexible subjects (~25 percent) (19). Thus, rotation permits interfascicle (and interfibre) translation. As there has been reported to be little difference between flexible and inflexible individuals in muscle lengthening and changes in fascicle length (series elastic component), the significant differences in fascicle rotation suggest a substantial difference in the parallel elastic component as well as the previously reported greater stretch tolerances of flexible individuals. The role of myofascia with muscle force and flexibility has become a much more involved debate. Deep fascia is also considered a key element to transmit load in parallel bypassing the joints, transmitting around 30 percent of the force generated during a muscular contraction.

Acute Morphological Proprioceptive Neuromuscular Facilitation (PNF) Stretching Mechanisms

Similar morphological mechanisms might be expected with PNF stretching. Static stretching is reported to be more effective in decreasing muscle stiffness (90), whereas PNF is more efficient in reducing both muscle and tendon stiffness (60). Perhaps, as PNF is reported to increase stretch tolerance more than static stretching (90), the higher stretch tolerance may provide greater stress upon both the muscle and the tendon. It might be rationalized that the contractions employed with contract-relax (CR) and contract-relax-agonist-contract (CRAC) PNF stretching provides a mental distraction (8), permitting a greater pain or stretch tolerance. As mentioned previously, the literature is conflicted as to whether PNF actually provides greater (39, 42, 46, 92), similar (28, 69) or trivial (85), increases in ROM compared with static stretching or even less flexibility than static stretching (111).

Acute Morphological Dynamic Stretching Mechanisms

As dynamic stretching uses repeated cyclical loading and unloading of the involved muscles, usually for a few minutes (44), these muscular contractions should induce shear stresses between muscle fibres and increase muscle temperature decreasing viscosity (thixotropy). In animal models, repeated lengthening and shortening of a muscle has increased muscle extensibility (88). There is one human dynamic stretching study indicating passive muscle stiffness reductions with increased ROM (55).

Dynamic, ballistic stretching has often not been advocated for efficiently improving ROM. Higher velocity dynamic stretching (ballistic) can weaken tissue (116, 117) by producing greater tensile forces over a brief duration (110). These high forces within a short time period do not allow for stress relaxation or creep to occur. Stress relaxation and creep refer to the reduction

in tension and tissue lengthening, respectively, that occur when a tissue is held in a lengthened position for an extended period of time (3). Hence, these mechanisms may explain why ballistic stretching is often found not to provide substantial increases in ROM and the fear that ballistic stretching is more likely to lead to injuries especially with muscle that is not warmed up. Unfortunately, there is limited literature rationalizing the ROM morphological mechanisms for dynamic stretching.

Summary

Increased flexibility can be achieved even without stretching. Increasing tissue (muscles and tendons) temperatures elicit thixotropic effects, which decrease tissue visco-elasticity. Static and PNF stretching can activate a number of inhibitory reflexes. Prolonged static modes of stretching can disfacilitate spindle reflexes as well as inducing pre- and post-synaptic inhibition of the afferents. Specifically, the activity (discharge frequency) of intrafusal, muscle spindle, nuclear chain (detects extent of stretch) and nuclear bag (detects the extent and rate of stretch) fibres diminish with the static modes of stretching. Although GTO are predominately inhibitory, they do not respond strongly to stretch and any extent of inhibition from GTO subside almost immediately following the stretch. Furthermore, Renshaw cells also do not play a substantial inhibitory role with static stretching. However, extensive static stretching can activate cutaneous afferent inhibition. Active static and PNF stretching, which involve the contraction of the antagonist muscle, could initiate the contribution of reciprocal inhibition. PNF stretching also uses many of these inhibitory mechanisms, whereas dynamic stretching tends to excite rather than inhibit these reflexes. Hence, the stretch tolerance theory can also help explain increased ROM as the individual accommodates the discomfort or pain associated with stretching and can push themselves past their prior ROM.

Morphological considerations include the skeletal configuration and alignment which cannot be modified with stretching. Ligaments which help secure bone to bone (joint stability) are primarily inelastic and avascular and thus are also resistant to elastic or plastic elongation by stretching except with the intense flexibility training routines of extreme athletes like gymnasts, dancers, figure skaters and others. Excessive fat can impede joint ROM. Nerves can acutely elongate approximately 6–20 percent of their resting length but thereafter they are susceptible to injury. Highly hypertrophied muscle can also provide some ROM restrictions, however the great extensibility of muscle puts into dispute the old theory of muscle boundness. Myofibrils can elongate double their resting length mainly due to the protein titin. Muscle is more compliant than tendons, with tendons accounting for less than half of the musculotendinous unit change. Muscle extensibility can also be altered by changes in fascicle angle (trivial), fascicle rotation (minor), and fascicle elongation (substantial). However, stretch tolerance may provide a greater impetus to enhanced ROM than decreased musculotendinous stiffness.

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